This operator can be obtained from the joint state by taking the *partial trace* over B, i.e. by taking the sum

$$\hat{\rho}_A = \sum_m {}_B\langle m| \Big(|\psi\rangle_{AB}\langle\psi| \Big) |m\rangle_B = \mathrm{Tr}_B \Big[|\psi\rangle_{AB}\langle\psi| \Big]. \tag{2.67}$$

Note that $\hat{\rho}_A$ is independent of the choice of bases for A and B (just like the full trace is basis independent).

Since (2.65) holds for *any* operator on A, it follows that all the information which can be learned by observing subsystem A alone is encoded in $\hat{\rho}_A$. We call $\hat{\rho}_A$ the *reduced state* of system A. Similarly, we can define a reduced state $\hat{\rho}_B = \mathrm{Tr}_A[|\psi\rangle_{AB}\langle\psi|]$ for system B, which encodes all that can be learned given acces to B alone. In general, the information contained in $\hat{\rho}_A$ and $\hat{\rho}_B$ together is not equivalent to that of the joint state $|\psi\rangle_{AB}$, because information about the correlations between A and B is lost if only one or the other subsystem is accessible. Several different joint states $|\psi\rangle_{AB}$ may be compatible with the same pair of reduced states $\hat{\rho}_A$, $\hat{\rho}_B$.

Example 2.3: Reduced states of entangled qubits

Consider two entangled two-qubit states

$$|\phi\rangle = \frac{1}{\sqrt{2}}(|0,0\rangle_{AB} + |1,1\rangle_{AB}), \tag{2.68}$$

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0,1\rangle_{AB} + |1,0\rangle_{AB}). \tag{2.69}$$

These states are not the same, in fact they are orthogonal $\langle\phi|\psi\rangle = 0$. They also encode different correlations between the subsystems. If both qubits are measured in the computational basis, for $|\phi\rangle$ the outcomes on A and B will also be identical while for $|\psi\rangle$ they will always be opposite. However, it is easy to see that the reduced states are the same

$$\hat{\rho}_A = \mathrm{Tr}_B|\phi\rangle\langle\phi| = \mathrm{Tr}_B|\psi\rangle\langle\psi| = \frac{1}{2}\mathbb{1}_A, \tag{2.70}$$

and similarly for B.

Note that the reduced state is *not* a ray in Hilbert space and thus not a pure state. Rather, it is a density operator.

2.3.4 Density operators

The reduced state (2.66) has unit trace $\mathrm{Tr}\hat{\rho}_A = \sum_{n,m} |\alpha_{nm}|^2 = 1$ (assuming that the joint state is normalised), it is hermitian $\hat{\rho}_A^\dagger = \hat{\rho}_A$, and it is positive semi-definite (the proof is left as Prob. 2.11). In fact, any operator with these properties can be understood as the reduced state of some larger system.

A *density operator* (or *density matrix*) is any operator $\hat{\rho}$ on a complex Hilbert space with the properties

$$\text{(i)} \quad \mathrm{Tr}[\hat{\rho}] = 1 \qquad \text{(unit trace).} \tag{2.71}$$

$$\text{(ii)} \quad \hat{\rho} \geq 0 \qquad \text{(positive semidefiniteness).} \tag{2.72}$$

Property (ii), that $\hat{\rho}$ is non-negative, implies that $\hat{\rho}$ is also Hermitian $\hat{\rho}^\dagger = \hat{\rho}$ (this holds for complex Hilbert spaces). Any density operator for a system S can be understood as the reduced state of S corresponding to some joint state $|\psi\rangle_{SS'}$ on a on larger system with Hilbert space $\mathcal{H}_S \otimes \mathcal{H}_{S'}$. If the state of S happens to be uncorrelated to anything outside S, then the density operator is a projector

$$\hat{\rho} = |\psi\rangle\langle\psi|. \tag{2.73}$$

In this case $\mathrm{Tr}[\hat{\rho}^2] = 1$, the density operator encodes the same information as the state vector $|\psi\rangle$, and we say that the state is *pure*. Otherwise, $\mathrm{Tr}[\hat{\rho}^2] < 1$ and we say that $\hat{\rho}$ is a *mixed* state. The quantity $\mathrm{Tr}[\hat{\rho}^2]$ is called the *purity* of $\hat{\rho}$. Note that $\frac{1}{d} \leq \mathrm{Tr}[\hat{\rho}^2] \leq 1$ where d is the dimension of the system.

Ensemble interpretation

Because $\hat{\rho}$ is Hermitian it can be diagonalised with real eigenvalues

$$\hat{\rho} = \sum_n p_n |n\rangle\langle n|, \tag{2.74}$$

in some basis $\{|n\rangle\}$. Furthermore, because $\hat{\rho}$ is positive semidefinite and has unit trace, the coefficients fulfill $p_n \geq 0$ and $\sum_n p_n = 1$ and thus form a probability distribution. It is therefore possible to interpret $\hat{\rho}$ as representing a situation where the exact state of the system is not known but it is known to be in each state $|n\rangle$ with probability p_n (equivalently, the state is drawn from an ensemble in which each $|n\rangle$ occurs with frequency p_n). That is, we have incomplete knowledge and $\hat{\rho}$ encodes our best description of the system state given this knowledge. The mixedness of $\hat{\rho}$ is due to our ignorance.

The Bloch ball

Like for pure states, mixed states of qubits also have a very useful, intuitive representation in terms of the Bloch ball. Because the three Pauli matrices together with the identity span the space of all 2x2 Hermitian matrices, any qubit density matrix can be expressed as

$$\hat{\rho} = \frac{1}{2}(\mathbb{1} + \mathbf{u} \cdot \hat{\boldsymbol{\sigma}}), \tag{2.75}$$

where $\mathbf{u}$ is a three-dimensional vector with norm $|\mathbf{u}| \leq 1$. For pure states, $|\mathbf{u}| = 1$, while for mixed states, $|\mathbf{u}| < 1$. Referring again to Fig. 2.1, this means that pure qubit states correspond to points on the surface of the unit sphere while mixed states correspond to points inside the ball. The center of the Bloch sphere corresponds to $\mathbf{u} = \mathbf{0}$, i.e.

$$\hat{\rho} = \frac{1}{2}\mathbb{1}, \tag{2.76}$$

which is the *maximaly mixed state*. The components of the Bloch vector can be computed as

$$u_x = \mathrm{Tr}[\hat{\rho}\hat{\sigma}_x], \qquad u_y = \mathrm{Tr}[\hat{\rho}\hat{\sigma}_y], \qquad u_z = \mathrm{Tr}[\hat{\rho}\hat{\sigma}_z], \tag{2.77}$$

as can be seen by using the properties of the Pauli operators.

Example 2.4: Probabilistic qubit source

Consider a source emitting qubits. The exact details of the source are unknown, but it is known to emit the two states

$$|\psi_\pm\rangle = \cos(\theta)|0\rangle \pm \sin(\theta)|1\rangle \tag{2.78}$$

with equal probabilities. What is the density matrix describing the source output? And what is its Bloch vector?

We can understand the source as an ensemble in which each $|\psi_\pm\rangle$ occurs with probability $\frac{1}{2}$. The density matrix is

$$\hat{\rho} = \frac{1}{2}|\psi_+\rangle\langle\psi_+| + \frac{1}{2}|\psi_-\rangle\langle\psi_-| = \cos^2(\theta)|0\rangle\langle 0| + \sin^2(\theta)|0\rangle\langle 0|. \tag{2.79}$$

Using the trigonometric identities $\cos^2(\theta) = \frac{1}{2}[1+\cos(2\theta)]$ and $\sin^2(\theta) = \frac{1}{2}[1-\cos(2\theta)]$, this matrix can be expressed as

$$\hat{\rho} = \frac{1}{2}(\mathbb{1} + \cos(2\theta)\hat{\sigma}_z), \tag{2.80}$$

so the Bloch vector is $\mathbf{u} = (0, 0, \cos(2\theta))$ along the z-axis of the Bloch sphere.

Von Neumann entropy

Another measure of the mixedness of a quantum state (in addition to the purity) is the von Neumann entropy, defined by

$$S(\hat{\rho}) = \mathrm{Tr}[\hat{\rho}\log(\hat{\rho})]. \tag{2.81}$$

This generalises the Shannon entropy for classical random variables to quantum states. In particular, evaluating S in a basis where $\hat{\rho}$ is diagonal, we get

$$S(\hat{\rho}) = \sum_n p_n \log(p_n) = H(\{p_n\}), \tag{2.82}$$

where p_n are the eigenvalues of $\hat{\rho}$, which define a probability distribution, and $H(\{p_n\})$ is the Shannon entropy of this distribution. Note that there is some freedom in the choice of base for the logarithm. If we wish to measure information in bits, we should choose base 2, i.e. $\log_2$.

The von Neumann entropy vanishes for pure states $S(\hat{\rho}) = 0$ and it is maximised for the maximally mixed state $S(\frac{1}{d}\mathbb{1}) = \log(d)$, where d is the system dimension. In addition, it is concave, i.e. for a set of density operators $\{\hat{\rho}_k\}$ and numbers $p_k \geq 0$, $\sum_k p_k = 1$

$$S(\sum_k p_k\hat{\rho}_k) \geq \sum_k p_k S(\hat{\rho}_k), \tag{2.83}$$

it is additive for independent systems

$$S(\hat{\rho}_A \otimes \hat{\rho}_B) = S(\hat{\rho}_A) + S(\hat{\rho}_B), \tag{2.84}$$

and it is subadditive for composite systems

$$S(\hat{\rho}_{AB}) \leq S(\hat{\rho}_A) + S(\hat{\rho}_B). \tag{2.85}$$